

Copy everything below the line into a `.md` file — it includes the cover page, executive summary, and the complete SRD, ready for PDF conversion.

SHRAMSETU

Software Requirements Document (SRD)

Project	ShramSetu — Trust-First Blue-Collar Hiring Platform
Prepared for	Blue Workforce Connect '26 · Wooble Hackathon
Version	1.0 — Frozen Scope
Team	[Team Name] — [Member Names]
Date	[Date]
Status	Approved for build · Change-controlled

Executive Summary

Millions of India's blue-collar workers — electricians, plumbers, machine operators, welders, delivery executives — find work through referrals, WhatsApp groups, and luck. They have no resumes, no verified digital identity, and no platform built for how they actually use technology: voice, mobile, and regional languages. Employers face the mirror problem: hiring is slow, fragmented, and dominated by unverified skill claims and no-shows.

ShramSetu (श्रमसेतु — "bridge of labour") is a full-stack recruitment platform that solves both sides with one spine: **trusted digital identities**. Unlike a conventional job portal, ShramSetu is a *trust system with a job board attached*.

Five pillars:

- Voice-First Kaam Profile** — a worker builds a complete, structured profile in under 3 minutes by *speaking* in English, Hindi, or Telugu; AI extracts and pre-fills, the human confirms.
- Skill Passport (Two-Sided Trust)** — tiered verification (New → ID Verified → Skill Verified → Top Pro) for workers *and* employers, with a visible approval mechanism.
- Explainable SmartMatch** — every candidate-job match shows a 0-100 score *and its reasoning* ("Why 87: 5/6 skills, 4 km, wage aligned").

4. **Speed Engine** — urgent jobs, an "Available Today" worker toggle, and a Kanban hiring pipeline built to crush time-to-hire.
5. **Kaam Card** — a public, WhatsApp-shareable skill card, turning the brief's "WhatsApp groups" weakness into a distribution channel.

Stack: Next.js + TypeScript + Supabase (Postgres, Auth, RLS, Storage), deployed on Vercel. AI is used strictly as a product component (speech-to-text, extraction, description generation, OCR pre-check) with human-in-the-loop confirmation at every step.

North-star metric demonstrated on camera: average time-to-hire in hours.

This SRD specifies all functional and non-functional requirements with IDs, priorities, and acceptance criteria; the data model; API contract; matching and trust algorithms; security plan; execution gates; test plan; golden demo path; and deliverables checklist — every element traceable to the hackathon's five evaluation criteria.

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1. Document Control

How to use this document: This SRD is the team's contract. Every requirement has an ID, a priority, and acceptance criteria. Reference IDs in commits(`feat(WRK-03): voice input field`). Any new feature idea mid-hackathon must displace an equal-priority item — this is the change-control policy that prevents scope creep.

Field	Value
Product name	ShramSetu (alternatives: KaamPass, SkillSetu)
Tagline	<i>A bridge between India's skilled hands and honest work.</i>
Team size	2-4 (confirm 2 minimum per FAQ before deadline)
Timeline	48 hours to code freeze, +24h for deliverables
Deployment target	Vercel (frontend + API) / Supabase (DB, Auth, Storage)
North-star metric	Average time-to-hire (hours)

1.1 Traceability: Rubric → Requirements → Demo Moments

Criterion	Weight	Covered by modules	Camera moment
Functionality & Completeness	30%	AUTH, WRK, EMP, PIP, DSH	Full loop: post → match → apply → hire in one take
Technical Implementation	25%	Schema, RLS, API, algorithms	Architecture slide + ER diagram in docs
UX & Innovation	20%	AI, VER, PUB, I18N	Voice-built profile + Skill Passport badge appearing live
Business Impact	15%	Strategy §2, roadmap	Impact slide + realistic wage/geography data
Presentation & Docs	10%	§16-18 deliverables	6:30 video, rehearsed once

2. Product Strategy

2.1 Core Thesis

Every competing team will build a job portal. ShramSetu is a **trust system with a job board attached**. The brief's exact phrase — *"trusted digital identities"* — is the product's spine, not a feature.

2.2 The Three Structural Insights (design constraints, not features)

1. **Workers have no resumes.** Profile creation must be voice-first, icon-driven, and sub-3-minutes.
2. **Workers live on WhatsApp and phones.** Mobile-first, shareable output (Kaam Card), vernacular (EN/HI/TE – Telugu mandatory).
3. **Employers' pain is trust + speed, not candidate supply.** Every employer flow optimizes time-to-hire.

2.3 Product Pillars

Pillar	Definition	Differentiator
Voice-First Kaam Profile	Profile built by speaking; AI structures it	Solves the no-resume problem; the demo's opening spectacle
Skill Passport (Two-Sided Trust)	Tiered verification for workers and employers	Two-sided trust protects workers from scam jobs
Explainable SmartMatch	Every match shows its reasoning	Explainable beats black-box in evaluation
Speed Engine	Urgent jobs + "Available Today" toggle + Kanban pipeline	Attacks the foreman's real problem: someone who shows up <i>today</i>
Kaam Card	Public shareable skill card for WhatsApp	Virality loop; references the brief's own "WhatsApp groups" line

2.4 Personas

Persona	Profile	Primary needs	Success looks like
Ravi , 28	Electrician, Bhimavaram, 8 yrs exp., Android, Telugu-first, no resume	Find daily-wage work nearby; prove skill without paperwork	Profile made in 3 min by voice; hired in 2 days
Priya , 34	HR coordinator, manufacturing unit, Autonagar Vijayawada	4 fitters this week; verified skills, low no-show risk	4 shortlisted in 20 min; time-to-hire < 48h
Admin	Platform verifier (internal)	Clear verification queue fast and fairly	Certificate approved → badge live in one click

3. Scope: MoSCoW

Priority	Items
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Priority	Items
MUST (ship or fail)	Auth + roles; worker profile (form mode); job CRUD; job feed + filters; one-tap apply; candidate search; pipeline Kanban + status updates; worker application tracker; basic employer dashboard; EN/HI/TE i18n; deployment live from Hour 2
SHOULD (the winning layer)	Voice profile builder; verification module + admin queue; employer verification; explainable match score; Kaam Card public page; "Available Today" toggle; urgent jobs; profile-strength meter
COULD (only if ahead)	Certificate OCR pre-check; endorsements; post-hire ratings; semantic match bonus; PWA manifest
WON'T (roadmap only)	Chat, payments, real SMS/WhatsApp API, native apps, DigiLocker integration (roadmap only), background-check APIs, multi-city ops logic

4. Functional Requirements

Priorities: **M** = Must, **S** = Should, **C** = Could. Each requirement lists its acceptance criterion (AC).

Module AUTH – Identity & Access

ID	Requirement	Pri	Acceptance Criteria
AUTH-01	Email magic-link authentication; no passwords	M	User receives link, clicks, is signed in < 30s
AUTH-02	Role selection (Worker / Employer) at first login	M	After role choice, user lands on role-correct home
AUTH-03	Role-based route guards (middleware)	M	Worker cannot open <code>/employer/</code> ; redirect to own home
AUTH-04	Session persistence + logout	M	Refresh keeps session; logout clears it
AUTH-05	Row Level Security on all tables (§10)	M	Worker A cannot read/write Worker B's profile via API

Module WRK – Worker Portal

ID	Requirement	Pri	Acceptance Criteria
WRK-01	3-step onboarding: (1) trade from icon grid (≥ 10 trades), (2) details form, (3) wage/radius/shift prefs	M	Complete in < 3 min; progress indicator visible

ID	Requirement	Pri	Acceptance Criteria
WRK-02	Profile fields: trade, years exp., city + lat/lng, wage range, shift, languages, bio, photo (optional)	M	All save and render on profile
WRK-03	Voice input mode: mic → speech-to-text (Web Speech API, hi-IN/te-IN/en-IN) → LLM extraction → pre-filled form → user confirms/edits before save	S	Dictated trade/exp/wage/bio populate fields; confirmation step always shown
WRK-04	Profile-strength meter mapped to Skill Passport	S	Meter % updates on each completed action
WRK-05	Job feed: cards with wage/day, distance, trade icon, employer verification badge, urgent ribbon; filters: trade, distance, wage, shift	M	Filters return correct results; feed < 1.5s
WRK-06	One-tap Apply (profile is the application)	M	Apply creates application row; button becomes "Applied"
WRK-07	Application tracker: timeline Applied → Shortlisted → Interview → Offer → Hired with timestamps	M	Status changes reflect within 5s
WRK-08	"Available Today" toggle	S	Filterable by employers (EMP-05)
WRK-09	WhatsApp share button on each job card (wa.me deep link)	S	Opens WhatsApp with job title + link
WRK-10	In-app notifications: status changes, new matching job	S	Badge count + list view

Module EMP – Employer Portal

ID	Requirement	Pri	Acceptance Criteria
EMP-01	Job post form: trade, title, headcount, wage range, location, start date, shift, urgency flag	M	Post live in feed < 10s
EMP-02	AI job-description generator from structured fields (editable before post)	S	Readable 3–4 sentence description in < 5s
EMP-03	Candidate search: filters – trade, experience, distance, trust tier, wage, availability, language	M	Results ranked by match score desc.
EMP-04	Candidate card shows match score + top reason; profile view shows Skill Passport, skills, experience, wage, distance	M	All data renders; verification badges visible

ID	Requirement	Pri	Acceptance Criteria
EMP-05	Candidate search honors "Available Today" + urgency	S	Urgent job search sorts available workers first
EMP-06	Pipeline Kanban: Applied → Shortlisted → Interview → Offer → Hired; drag or action buttons; bulk shortlist	M	Status change persists + notifies worker
EMP-07	Hire action sets hired_at timestamp; prompts optional endorsement	S	Endorsement appears on worker passport

Module VER – Verification & Skill Passport (Two-Sided)

ID	Requirement	Pri	Acceptance Criteria
VER-01	Worker uploads ID / skill certificate (ITI, license) → private storage bucket, signed URLs only	M	File stored; publicly inaccessible
VER-02	Verification statuses: pending → approved / rejected (with note)	M	Transitions auditable (reviewed_at, reviewer)
VER-03	Approval upgrades worker's Trust Tier; badge appears instantly	M	Badge visible on next page load
VER-04	Employer verification: company doc upload → "Verified Employer" badge	S	Same pipeline as VER-01-03 (module reuse)
VER-05	OCR pre-check extracts name/cert type for admin speedup	C	Extracted fields shown alongside document in queue
VER-06	PII minimization: store status only; masked IDs; never store full ID numbers in plaintext	M	No raw ID number retrievable from DB or logs
VER-07	Roadmap note (docs only): DigiLocker + NSDC registry integration	—	Stated in technical doc + video

Module MAT – Matching Engine

ID	Requirement	Pri	Acceptance Criteria
MAT-01	Match score 0–100 computed per §8.1; cached in <code>match_scores</code>	M	Score appears on feed and candidate search
MAT-02	Explanation endpoint: top 3 contributing factors, plain language	S	"Why 87" panel renders in both portals
MAT-03	Skills modeled as structured taxonomy (not free text)	M	Match quality demonstrably depends on taxonomy

ID	Requirement	Pri	Acceptance Criteria
MAT-04	Semantic similarity bonus (embeddings), capped at +5	C	Bonus visible in score breakdown

Module DSH – Dashboards

ID	Requirement	Pri	Acceptance Criteria
DSH-01	Employer dashboard: active jobs, new applicants today, funnel (views→applies→shortlisted→hired), avg time-to-hire (headline)	M	All metrics from real seeded data; funnel charted
DSH-02	Worker dashboard: applications in review, profile views count, recommended jobs	M	"3 employers viewed your profile" insights
DSH-03	Per-job detail: applicants by stage, views, match-score distribution	S	Opens from dashboard click

Module I18N – Language & Accessibility

ID	Requirement	Pri	Acceptance Criteria
I18N-01	Full UI toggle: English / हिंदी / తెలుగు; all strings externalized	M	Toggle switches every screen; preference persisted
I18N-02	Icon + text labels everywhere (no icon-only)	M	Spot-check passes on all primary screens
I18N-03	≥ 48px touch targets, high contrast, font scaling	M	Lighthouse accessibility ≥ 90
I18N-04	Mobile-first worker flows; desktop-first employer flows	M	Worker flow flawless at 375px width

Module ADM – Admin Console

ID	Requirement	Pri	Acceptance Criteria
ADM-01	Admin-only verification queue: doc preview, extracted fields, approve/reject + note	S	Approve → VER-03 triggers; non-admin blocked
ADM-02	Platform stats strip (users, jobs, hires)	C	Renders on admin home

Module PUB – Kaam Card (Public Shareable Identity)

ID	Requirement	Pri	Acceptance Criteria
PUB-01	Public page <code>/c/{worker}</code> : trade, passport tier, skills, experience, wage, city – no PII beyond first name	S	Opens logged-out; contact gated behind platform
PUB-02	WhatsApp share with OG preview image	S	Link preview renders card details
PUB-03	Privacy: worker can disable public card	C	Toggle removes public access

5. Non-Functional Requirements

ID	Category	Requirement	Target
NFR-01	Performance	Feed & search p95 response	< 1.5s
NFR-02	Performance	Page load on throttled low-end Android profile	Usable < 3s
NFR-03	Security	RLS on every table; no client-side trust	All policies tested (§10)
NFR-04	Security	Input validation (zod) on all forms & API routes	Invalid input rejected with message
NFR-05	Security	Private storage bucket, signed URLs, file type/size limits (PDF/JPG, ≤ 5MB)	Verified by direct URL test
NFR-06	Privacy	PII minimization; masked IDs; no sensitive data in logs	Code review pass
NFR-07	Reliability	Deployed from Hour 2; final deploy ≥ 3h before deadline	Deploy history on Vercel
NFR-08	Scalability	Indexed queries; paginated feeds; matching extractable to background jobs	Documented in tech doc
NFR-09	Maintainability	TypeScript strict, typed schema exports, consistent structure	Repo passes team review
NFR-10	Explainability	Every AI feature has human-in-the-loop confirmation	Voice + verification flows show confirm steps

6. Data Model

users	id, auth_id, role(worker employer admin), email, created_at
worker_profiles	id, user_id→users, full_name, trade_id→skills, years_exp, city, lat, lng, wage_min, wage_max, shift_pref, languages[], bio, photo_url, available_today(bool), trust_tier, trust_score, passport_public(bool), created_at
skills	id, name_en, name_hi, name_te, category (taxonomy)
worker_skills	worker_id, skill_id, proficiency(1-5)
employer_profiles	id, user_id→users, company_name, industry, city, is_verified, created_at
jobs	id, employer_id→employer_profiles, title, trade_id→skills, headcount, wage_min, wage_max, city, lat, lng, shift, is_urgent, status(open closed), description, posted_by, created_at
job_skills	job_id, skill_id, required(bool)
applications	id, job_id, worker_id, status(applied shortlisted interview offer hired rejected), applied_at, shortlisted_at, interview_at, offer_at, hired_at ← powers time-to-hire
verification_documents	id, owner_user_id, doc_type(id skill_cert company), file_url, extracted_json, status, reviewer_note, reviewed_at, reviewed_by
endorsements	id, worker_id, employer_id, skill_id, comment, created_at
ratings	id, application_id, rater_id, ratee_id, score(1-5), comment
match_scores	job_id, worker_id, score, breakdown_json, computed_at
notifications	id, user_id, type, payload_json, read(bool), created_at

ER diagram: render in Excalidraw/draw.io for the technical doc. **Indexes:** jobs(trade_id, city) , jobs(status, is_urgent) , worker_profiles(trade_id, available_today) , applications(status) , match_scores(job_id, score desc) .

7. API Contract (primary routes)

Endpoint	Method	Auth	Purpose
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Endpoint	Method	Auth	Purpose
/api/jobs	GET	any	Feed w/ filters + pagination
/api/jobs	POST	employer	Create job
/api/jobs/:id	PATCH	owner	Close/update
/api/applications	POST	worker	One-tap apply
/api/applications/:id	PATCH	employer (owner)	Status transition + notification + stage timestamp
/api/candidates/search	GET	employer	Filtered, ranked by cached match score
/api/verifications	POST	worker/employer	Upload doc → pending
/api/admin/verifications/:id	PATCH	admin	Approve/reject + note
/api/match/explain	GET	any-auth	Score breakdown for job × worker
/api/ai/voice-profile	POST	worker	Transcript → structured fields (LLM)
/api/ai/job-description	POST	employer	Structured fields → description (LLM)
/api/dashboard/employer	GET	employer	Funnel + time-to-hire aggregates
/api/dashboard/worker	GET	worker	Tracker + insights
/c/[slug]	GET	public	Kaam Card (no PII)

All request bodies validated with zod; all handlers enforce session role before touching data.

8. Algorithm Specifications

8.1 Match Score (0–100)

$$\text{score} = 100 \times (0.35 \cdot S + 0.25 \cdot D + 0.15 \cdot E + 0.15 \cdot W + 0.10 \cdot T) + \text{bonus}$$

$$S \text{ skill overlap} = |\text{worker_skills} \cap \text{job_skills}| / |\text{job_skills}|$$

$$D \text{ distance} = 1.0 \text{ if } \leq 5\text{km (haversine)}; \text{ linear decay to } 0 \text{ at worker's max radius}$$

$$E \text{ experience} = 1.0 \text{ if } \text{exp} \geq \text{required}; 0.7 \text{ if within } 1 \text{ yr below}; \text{ else } 0.3$$

$$W \text{ wage alignment} = 1.0 \text{ if expected within range}; 0.6 \text{ if } \leq +10\% \text{ over}; \text{ else}$$

0.2
T trust tier = New 0.2 | ID 0.5 | Skill 0.8 | Top Pro 1.0
bonus = +5 max, embedding similarity (COULD)

Explanation strings generated from the largest three contributing components. **Explainable > black box – say this in the video.**

8.2 Trust Score & Tiers

trust = 30 (base)
+ 20 (ID verified)
+ 10 × approved skill certificates (cap 30)
+ 5 × completed hires (cap 10)
+ 4 × employer endorsements (cap 12) → max 100
Tiers: 0-39 New | 40-59 ID Verified | 60-84 Skill Verified | 85+ Top Pro

Gaming resistance (docs note): endorsements exist only platform-mediated, post-hire.

8.3 Voice Profile Pipeline

Mic → Web Speech API (te-IN/hi-IN/en-IN) → transcript
→ LLM extraction → JSON {trade, years_exp, wage, bio, languages}
→ pre-fill form → USER CONFIRMS/EDITS → save (human-in-the-loop, always)
Fallback: standard form (voice is enhancement, never a blocker)

9. Architecture

Next.js 14 (App Router) + TypeScript + Tailwind + shadcn/ui + Supabase (Postgres, Auth, Storage, RLS) + Vercel.

Browser (worker: mobile / employer: desktop)
| RSC + client components
Next.js on Vercel — server routes (/api/*) — zod validation
| |
▼ ▼
Supabase Auth Supabase Postgres (RLS on all tables)
Supabase Storage (private, signed URLs)
External AI: Web Speech API (STT, free) + LLM API (extraction/description)

AI policy (per hackathon FAQ): AI APIs are product components; core (auth, RLS, pipeline, matching math) is team-written. Every module must be explainable line-by-line by any team member – enforced

10. Security & Privacy Plan

Control	Implementation
RLS policies	Workers: write own profile only. Employers: read applicants only via own jobs. Admin: verification tables. Everyone: read only open jobs.
Role guards	Middleware on <code>/employer/*</code> , <code>/admin/*</code>
Validation	zod schemas shared client/server
File uploads	Private bucket, signed URLs, type+size limits
PII	No raw ID numbers stored; masked display; statuses only
Transport	HTTPS everywhere (Vercel default)
Logging	No tokens/PII in logs

11. Screen Inventory

Route	User	Purpose
<code>/</code>	public	Landing: value prop, two CTA cards, language toggle
<code>/login</code>	all	Magic-link
<code>/onboarding/worker</code>	worker	3-step builder (voice-capable)
<code>/home</code>	worker	Feed + available-today toggle + notifications
<code>/profile</code>	worker	Passport, strength meter, skills, Kaam Card settings
<code>/jobs/:id</code> , <code>/applications</code>	worker	Detail + tracker timeline
<code>/employer/post</code> , <code>/employer/jobs</code>	employer	Post form (AI desc), own jobs
<code>/employer/candidates</code> , <code>/candidates/:id</code>	employer	Search + passport view
<code>/employer/pipeline</code>	employer	Kanban

Route	User	Purpose
/employer/dashboard	employer	Funnel + time-to-hire
/verify	worker/employer	Upload docs
/admin/verifications	admin	Queue
/c/[slug]	public	Kaam Card

12. Execution Plan (Gates, not hours)

Gate	When	Exit criteria (hard, binary)
G0 Setup	H+2	Repo, scaffold, Supabase connected, empty app live on Vercel , schema + seed script committed, i18n string sheet created, design tokens fixed
G1 Skeleton	H+10	Auth + role onboarding both roles; worker profile form saves; job CRUD live; deployed
G2 Core Loop	H+20	Feed + filters + apply + candidate search + Kanban + tracker + notifications work end-to-end as both roles on the live deployment
G3 Differentiators	H+30	Voice builder, verification + admin queue + badges, explainable match scores, Kaam Card, urgent/available – whichever are committed by freeze
G4 Freeze	H+34	Feature freeze. Seed rich data (20 workers, Telugu names, Bhimavaram/Rajahmundry/Vijayawada; 10 jobs, ₹700–1,100/day; applications across all stages). Empty/loading states. Mobile pass. Lighthouse pass.
G5 Ship	H+40	Final deploy + full regression (§13) + deliverables uploaded ≥ 3h before deadline. Stop coding.

Team lanes (3–4 members): L1 Product/Frontend (WRK, I18N) · L2 Backend/Data (AUTH, EMP, DSH, RLS) · L3 AI/Innovation (AI, VER, MAT, ADM) · L4 Deliverables/QA (§16, seeding, testing – rotates into build lanes). *Two-person team:* merge L1+L4, L2+L3; downgrade voice builder to one voice-input field + guided form.

13. Test Plan (run twice: at G2 and G5)

#	Test	Pass condition
T1	Worker magic-link signup → onboard → profile saved	< 3 min total

#	Test	Pass condition
T2	Voice dictation populates fields; user edit respected	Confirmed save
T3	Employer posts urgent job	Live in feed < 10s
T4	Worker feed filters by trade+distance+wage	Correct results
T5	One-tap apply	Application in tracker "Applied"
T6	Employer shortlists → interviews → hires	Worker tracker updates each stage; timestamps set
T7	Worker uploads cert; admin approves	Badge appears on candidate card
T8	Match explanation renders	"Why 87" breakdown correct per formula
T9	RLS probe: Worker B token cannot read Worker A profile	403/empty
T10	i18n toggle EN/Hi/TE across 5 core screens	All strings switch
T11	Worker flow at 375px width	No overflow, no dead taps
T12	Public Kaam Card logged-out; no PII beyond first name	Passes

14. Golden Demo Path (used for BOTH final QA and the video)

1. Ravi lands → language → తెలుగు → onboarding with **voice** → confirm → passport shown (New tier)
2. Upload ITI certificate → admin approves → **Skill Verified badge appears**
3. Feed in Telugu → urgent job → one-tap apply → tracker
4. Switch: Priya posts job (AI description) → candidate search ranked with match scores → "Why 87" → Ravi's passport → shortlist → hire
5. Dashboard: **time-to-hire headline** + funnel with seeded data
6. Share Ravi's **Kaam Card** to WhatsApp → OG preview
7. Architecture slide (30s) → impact + roadmap (DigiLocker, employer SaaS) → close on mission

15. AI-Assisted Development Playbook

Rules of engagement (non-negotiable, per hackathon FAQ):

1. AI accelerates; you own. Never merge code you can't explain.
2. Every AI-assisted module gets an entry in the **Explanation Log** (markdown file: module → 5-line plain explanation → one edge case → who can present it).
3. AI is a product component (voice, extraction, description), never the architecture.

Prompt templates (copy-adapt):

- **Schema/RLS:** "Here is my schema [paste]. Write Supabase RLS policies such that: workers write only their own profile; employers read applicants only via their own jobs; admins write verification tables; all authenticated users read open jobs. Output as SQL with a one-line rationale per policy."
- **Component scaffold:** "Create a Next.js client component: JobCard with props [list]. Props validated with zod. Tailwind + shadcn conventions. Touch targets $\geq 48\text{px}$. Then explain what each part does in 3 bullets."
- **Match function:** "Implement `computeMatch(worker, job)` in TypeScript per this formula [paste §8.1], returning score + breakdown object. Pure function, unit-testable, no DB calls."
- **LLM extraction:** "You are an information extraction service. Given this speech transcript [var], return STRICT JSON with keys trade, years_experience, wage_min, wage_max, bio, languages. Values must map to this taxonomy [paste]. If a field is absent, return null. No prose."
- **Seed data:** "Generate a Supabase seed script: 20 blue-collar workers in coastal Andhra Pradesh (Bhimavaram, Tadepalligudem, Rajahmundry, Vijayawada), realistic Telugu names, trades across electrician/plumber/welder/CNC/fitter, daily wages ₹600–1,400, varied trust tiers; 10 jobs from verified employers; 30 applications spread across pipeline stages with realistic timestamps."
- **i18n:** "Here are 40 English UI strings for a blue-collar hiring app [paste]. Produce a JSON file with en/hi/te keys. Keep register simple, respectful, jargon-free."

16. Deliverables & Submission Checklist

Deliverable	Spec	Owner
Web app	Full loop live, G5 passed	All
GitHub repo	README (banner, one-liner, live link, demo creds for worker+employer+admin , screenshots, 3-step setup), Explanation Log, ER diagram	L4
Demo video	6:30 target, 1080p, storyboard = §14, one rehearsal, rubric vocabulary spoken naturally	L4 + all
Technical documentation	§1–15 of this SRD realized: architecture, stack rationale, ER, API table, algorithms, security, AI ethics, scalability, limitations/roadmap	L4
Deployment link	Vercel, seeded, stable $\geq 3\text{h}$ pre-deadline	L2

Deliverable	Spec	Owner
Deck (optional, do it)	8 slides: problem → insight ("no resumes, no trust, no speed") → 3 pillars → demo shots → architecture → trust system → business model (worker-free; employer ₹2-5k/mo SaaS + credits + verification-as-a-service; moat = verified worker graph) → roadmap	L1

Final pre-submission checklist: full loop on live URL ✓ voice demo works twice ✓ badge flow on camera ✓ match explanation ✓ dashboard numbers real ✓ EN/HI/TE ✓ README creds ✓ video 6:30 ✓ uploaded ≥ 3h early ✓ team roster 2-4 confirmed ✓

17. Risk Register

Risk	Mitigation
Deploy breaks at deadline	Live from H+2; redeploy G2/G4; freeze G4
AI latency on camera	Precompute demo-path matches; extraction on short transcripts
Voice API fails on judge's browser	Graceful form fallback; demo on Chrome
RLS lockout mid-hack	Test both roles hourly after G1
Fake-looking data	Seeded Telugu-local script (§12 G4)
Video overruns	§14 storyboard is the clock; 6:30 hard cap
Scope creep	G4 freeze; new ideas go to "roadmap" slide, not codebase
Can't explain AI-assisted code	Explanation Log (§15), no exceptions

18. Definition of Done

ShramSetu v1.0 is done when: the **golden demo path (§14) runs end-to-end on the public deployment without error**, all MUST requirements pass their acceptance criteria, all 12 regression tests pass, every deliverable in §16 is uploaded to Wooble, and every team member can explain any module from the Explanation Log on request.

One-line constitution for the team: *We are not building a job portal; we are building the trust layer India's blue-collar workforce has never had — every requirement above exists to prove that on camera, in code, and in documentation.*